SIMATS ENGINEERING

ASSESSMENT TOOL 3 – MCQ ANSWER KEY

Course Name: Artificial Intelligence   Course Code: CSA17

CO2: Search and game-solving techniques – Greedy Search, A*, CSP, Minimax, Alpha-Beta pruning.

Q#	Answer	Correct Option	Why
1	**C**	Greedy Best-First Search	Chooses the node with the lowest h(n) only, ignoring path cost.
2	**C**	f(n) = g(n) + h(n)	A* combines actual cost so far with estimated remaining cost.
3	**B**	Admissibility	An admissible heuristic never overestimates true cost, ensuring A* optimality.
4	**B**	Hill Climbing	A local search method that iteratively improves the current solution.
5	**B**	Minimax Algorithm	Standard decision procedure for two-player adversarial games.
6	**B**	Greedy Best-First Search	Selects based on estimated distance only, not accumulated cost.
7	**B**	A* Search	Combines distance travelled with estimated remaining distance.
8	**B**	Constraint Satisfaction Problem	Variables (exams), domains (slots/rooms), constraints (no overlap, room limits).
9	**C**	Minimax Algorithm	Anticipates opponent responses and assigns utility values to states.
10	**B**	Online Search Agent	Acts and senses in an unknown environment, updating knowledge as it goes.
11	**B**	It ensures optimality	Admissibility guarantees A* never overestimates, so it finds the optimal path.
12	**B**	Reduces number of nodes evaluated	Alpha-Beta skips branches that cannot affect the final decision.
13	**C**	Getting stuck in local maxima	Hill Climbing has no lookahead, so it can stop at a local optimum.
14	**C**	Constraint satisfaction	A valid solution must satisfy all constraints over all variables.
15	**B**	Estimate utility of a state	Evaluation functions approximate how good a non-terminal state is.

16	**A**	12	f(n) = g(n) + h(n) = 7 + 5 = 12.
17	**B**	A* becomes non-optimal	An overestimating (inadmissible) heuristic breaks A*'s optimality guarantee.
18	**C**	31	Full binary tree of depth 4: 2^0+2^1+2^2+2^3+2^4 = 31 nodes.
19	**B**	27	Leaf nodes = branching factor^depth = 3^3 = 27.
20	**B**	O(b^(d/2))	Best-case Alpha-Beta pruning halves the effective search depth.